

Mathematical Foundations of Machine Learning

Assignment 10 Solutions

Question 1 (15 points)

Let X be a continuous random variable with pdf

$$f_X(x) = \begin{cases} 1 - |x|, & -1 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) (5 points) Compute $E[|X|]$.
- (b) (5 points) Compute $E[X^2]$.
- (c) (5 points) Compute $\text{Var}(X)$.

Solution.

- (a) Since the density is symmetric about 0,

$$E[|X|] = 2 \int_0^1 x(1 - x) dx.$$

Therefore,

$$\begin{aligned} E[|X|] &= 2 \int_0^1 (x - x^2) dx \\ &= 2 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 \\ &= 2 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3}. \end{aligned}$$

Hence

$$\boxed{E[|X|] = \frac{1}{3}.$$

- (b) Again using symmetry,

$$E[X^2] = 2 \int_0^1 x^2(1 - x) dx.$$

Therefore,

$$\begin{aligned} E[X^2] &= 2 \int_0^1 (x^2 - x^3) dx \\ &= 2 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 \\ &= 2 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{1}{6}. \end{aligned}$$

Hence

$$E[X^2] = \frac{1}{6}.$$

(c) Since the density is symmetric about 0, we have $E[X] = 0$. Therefore,

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$

Using part (b),

$$\text{Var}(X) = \frac{1}{6} - 0 = \frac{1}{6}.$$

Hence

$$\text{Var}(X) = \frac{1}{6}.$$

Question 2 (25 points)

Let (X, Y) be a pair of continuous random variables with joint pdf

$$f_{X,Y}(x, y) = \begin{cases} 2, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) (5 points) Compute the marginal pdfs $f_X(x)$ and $f_Y(y)$.
- (b) (5 points) Compute the conditional pdfs $f_{Y|X}(y|x)$ and $f_{X|Y}(x|y)$.
- (c) (5 points) Compute $E[X]$ and $E[Y]$.
- (d) (5 points) Compute $E[X^2]$, $E[Y^2]$, and $E[XY]$.
- (e) (5 points) Compute $\text{Var}(X)$, $\text{Var}(Y)$, $\text{Cov}(X, Y)$, and $\text{Corr}(X, Y)$.

Solution.

- (a) For $0 < x < 1$,

$$f_X(x) = \int_0^x 2 \, dy = 2x.$$

Thus

$$f_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

For $0 < y < 1$,

$$f_Y(y) = \int_y^1 2 \, dx = 2(1 - y).$$

Thus

$$f_Y(y) = \begin{cases} 2(1 - y), & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (b) For $0 < y < x < 1$,

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \frac{2}{2x} = \frac{1}{x}.$$

Hence

$$f_{Y|X}(y|x) = \begin{cases} \frac{1}{x}, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

For $0 < y < x < 1$,

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{2}{2(1-y)} = \frac{1}{1-y}.$$

Hence

$$f_{X|Y}(x|y) = \begin{cases} \frac{1}{1-y}, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(c) Using $f_X(x) = 2x$,

$$E[X] = \int_0^1 x(2x) dx = 2 \int_0^1 x^2 dx = \frac{2}{3}.$$

Using $f_Y(y) = 2(1-y)$,

$$E[Y] = \int_0^1 y \cdot 2(1-y) dy = 2 \int_0^1 (y - y^2) dy = \frac{1}{3}.$$

Hence

$$\boxed{E[X] = \frac{2}{3}, \quad E[Y] = \frac{1}{3}.}$$

(d) First,

$$E[X^2] = \int_0^1 x^2(2x) dx = 2 \int_0^1 x^3 dx = \frac{1}{2}.$$

Next,

$$E[Y^2] = \int_0^1 y^2 \cdot 2(1-y) dy = 2 \int_0^1 (y^2 - y^3) dy = \frac{1}{6}.$$

Also,

$$\begin{aligned} E[XY] &= \int_0^1 \int_0^x 2xy dy dx \\ &= \int_0^1 x [y^2]_0^x dx \\ &= \int_0^1 x^3 dx = \frac{1}{4}. \end{aligned}$$

Hence

$$\boxed{E[X^2] = \frac{1}{2}, \quad E[Y^2] = \frac{1}{6}, \quad E[XY] = \frac{1}{4}.}$$

(e) We compute

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18},$$

and

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{18}.$$

Also,

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = \frac{1}{4} - \frac{2}{3} \cdot \frac{1}{3} = \frac{1}{36}.$$

Therefore,

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{1/36}{\sqrt{(1/18)(1/18)}} = \frac{1}{2}.$$

Hence

$$\boxed{\text{Var}(X) = \frac{1}{18}, \quad \text{Var}(Y) = \frac{1}{18}, \quad \text{Cov}(X, Y) = \frac{1}{36}, \quad \text{Corr}(X, Y) = \frac{1}{2}.}$$

Question 3 (15 points)

Let (X, Y) be a pair of continuous random variables with joint pdf

$$f_{X,Y}(x, y) = \begin{cases} 4x, & 0 < x < 1, 0 < y < x^2, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) (5 points) Compute the marginal pdfs $f_X(x)$ and $f_Y(y)$.
- (b) (5 points) Compute the conditional mean $E[X|Y = y]$.
- (c) (5 points) Verify that $E[E[X|Y]] = E[X]$.

Solution.

- (a) For $0 < x < 1$,

$$f_X(x) = \int_0^{x^2} 4x \, dy = 4x^3.$$

Thus

$$f_X(x) = \begin{cases} 4x^3, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

For $0 < y < 1$, the condition $0 < y < x^2$ is equivalent to $\sqrt{y} < x < 1$. Hence

$$f_Y(y) = \int_{\sqrt{y}}^1 4x \, dx = 2 \left[x^2 \right]_{\sqrt{y}}^1 = 2(1 - y).$$

Therefore,

$$f_Y(y) = \begin{cases} 2(1 - y), & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (b) For $0 < y < 1$ and $\sqrt{y} < x < 1$,

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{4x}{2(1 - y)} = \frac{2x}{1 - y}.$$

Hence

$$\begin{aligned} E[X|Y = y] &= \int_{\sqrt{y}}^1 x f_{X|Y}(x|y) \, dx \\ &= \frac{2}{1 - y} \int_{\sqrt{y}}^1 x^2 \, dx \\ &= \frac{2}{1 - y} \left[\frac{x^3}{3} \right]_{\sqrt{y}}^1 \\ &= \frac{2(1 - y^{3/2})}{3(1 - y)}. \end{aligned}$$

Therefore,

$E[X Y = y] = \frac{2(1 - y^{3/2})}{3(1 - y)} = \frac{2(1 + \sqrt{y} + y)}{3(1 + \sqrt{y})}, \quad 0 < y < 1.$
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(c) From part (b),

$$E[X|Y = y] = \frac{2(1 - y^{3/2})}{3(1 - y)}, \quad 0 < y < 1,$$

and from part (a),

$$f_Y(y) = 2(1 - y), \quad 0 < y < 1.$$

Therefore,

$$\begin{aligned} E[E[X|Y]] &= \int_0^1 E[X|Y = y] f_Y(y) dy \\ &= \int_0^1 \frac{2(1 - y^{3/2})}{3(1 - y)} \cdot 2(1 - y) dy \\ &= \frac{4}{3} \int_0^1 (1 - y^{3/2}) dy \\ &= \frac{4}{3} \left[y - \frac{2}{5} y^{5/2} \right]_0^1 \\ &= \frac{4}{3} \left(1 - \frac{2}{5} \right) = \frac{4}{5}. \end{aligned}$$

Also, using $f_X(x) = 4x^3$ from part (a),

$$E[X] = \int_0^1 x \cdot 4x^3 dx = 4 \int_0^1 x^4 dx = \frac{4}{5}.$$

Hence

$$E[E[X|Y]] = E[X] = \frac{4}{5}.$$

Question 4 (15 points)

Let (X, Y, Z) be a triple of continuous random variables with joint pdf

$$f_{X,Y,Z}(x, y, z) = \begin{cases} 15xz, & 0 < y < x < z \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) (3 points) Compute the marginal pdf $f_Z(z)$.
- (b) (3 points) Compute the conditional joint pdf $f_{X,Y|Z}(x, y|z)$.
- (c) (4 points) Compute $E[XY|Z = z]$.
- (d) (5 points) Using your answer from part (a), determine a median of Z and the mode of Z . Briefly justify your answers.

Solution.

- (a) For $0 < z \leq 1$,

$$f_Z(z) = \int_0^z \int_0^x 15xz dy dx = 15z \int_0^z x^2 dx = 5z^4.$$

Thus

$$f_Z(z) = \begin{cases} 5z^4, & 0 < z \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(b) For $0 < z \leq 1$ and $0 < y < x < z$,

$$f_{X,Y|Z}(x,y|z) = \frac{f_{X,Y,Z}(x,y,z)}{f_Z(z)} = \frac{15xz}{5z^4} = \frac{3x}{z^3}.$$

Hence

$$f_{X,Y|Z}(x,y|z) = \begin{cases} \frac{3x}{z^3}, & 0 < y < x < z \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(c) Using the conditional joint pdf from part (b),

$$\begin{aligned} E[XY|Z=z] &= \int_0^z \int_0^x xy \frac{3x}{z^3} dy dx \\ &= \frac{3}{z^3} \int_0^z x^2 \left(\int_0^x y dy \right) dx \\ &= \frac{3}{2z^3} \int_0^z x^4 dx \\ &= \frac{3}{2z^3} \cdot \frac{z^5}{5} = \frac{3z^2}{10}. \end{aligned}$$

Hence

$$E[XY|Z=z] = \frac{3z^2}{10}, \quad 0 < z \leq 1.$$

(d) From part (a), the marginal pdf is

$$f_Z(z) = 5z^4, \quad 0 < z \leq 1.$$

Since $f_Z(z)$ is increasing on $(0, 1]$, its maximum occurs at the right endpoint. Hence the mode is

$$\boxed{1.}$$

For the median, first compute the cdf:

$$F_Z(z) = \int_0^z 5t^4 dt = z^5, \quad 0 < z \leq 1.$$

A median m satisfies $F_Z(m) = 1/2$, so

$$m^5 = \frac{1}{2}.$$

Therefore,

$$\boxed{m = 2^{-1/5}.}$$

Question 5 (10 points)

Let

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$$

be a random vector with

$$E[\mathbf{X}] = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \quad E[\mathbf{X}\mathbf{X}^\top] = \begin{bmatrix} 5 & 0 & 2 \\ 0 & 3 & -3 \\ 2 & -3 & 7 \end{bmatrix}.$$

(a) (6 points) Compute the covariance matrix $\text{Var}(\mathbf{X})$.

(b) (4 points) Compute $\text{Corr}(X_1, X_2)$.

Solution.

(a) We use

$$\text{Var}(\mathbf{X}) = E[\mathbf{X}\mathbf{X}^\top] - E[\mathbf{X}]E[\mathbf{X}]^\top.$$

First,

$$E[\mathbf{X}]E[\mathbf{X}]^\top = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix}.$$

Therefore,

$$\text{Var}(\mathbf{X}) = \begin{bmatrix} 5 & 0 & 2 \\ 0 & 3 & -3 \\ 2 & -3 & 7 \end{bmatrix} - \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}.$$

$$\boxed{\text{Var}(\mathbf{X}) = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}.$$

(b) From part (a),

$$\text{Var}(X_1) = 4, \quad \text{Var}(X_2) = 2, \quad \text{Cov}(X_1, X_2) = 1.$$

Therefore,

$$\text{Corr}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1)\text{Var}(X_2)}} = \frac{1}{\sqrt{4 \cdot 2}} = \frac{1}{2\sqrt{2}}.$$

Hence

$$\boxed{\text{Corr}(X_1, X_2) = \frac{1}{2\sqrt{2}}.}$$

Question 6 (20 points)

Let $\mathbf{X}$ be a random vector in $\mathbb{R}^n$ with finite covariance matrix $\text{Var}(\mathbf{X})$. Let

$$Y = \mathbf{a}^\top \mathbf{X} + b, \quad Z = \mathbf{c}^\top \mathbf{X} + d,$$

where $\mathbf{a}, \mathbf{c} \in \mathbb{R}^n$ and $b, d \in \mathbb{R}$ are constants.

(a) (8 points) Show that

$$\text{Var}(Y) = \mathbf{a}^\top \text{Var}(\mathbf{X}) \mathbf{a}, \quad \text{Var}(Z) = \mathbf{c}^\top \text{Var}(\mathbf{X}) \mathbf{c}.$$

(b) (6 points) Show that

$$\text{Cov}(Y, Z) = \mathbf{a}^\top \text{Var}(\mathbf{X}) \mathbf{c}.$$

(c) (6 points) Assume additionally that $\mathbf{a}^\top \text{Var}(\mathbf{X}) \mathbf{a} > 0$. If $\mathbf{c} = \lambda \mathbf{a}$ for some nonzero scalar λ , show that

$$\text{Corr}(Y, Z) = \begin{cases} 1, & \lambda > 0, \\ -1, & \lambda < 0. \end{cases}$$

Solution.

(a) Let $\boldsymbol{\mu} = E[\mathbf{X}]$. Then

$$Y - E[Y] = \mathbf{a}^\top \mathbf{X} + b - (\mathbf{a}^\top \boldsymbol{\mu} + b) = \mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu}).$$

Therefore,

$$\begin{aligned} \text{Var}(Y) &= E[(Y - E[Y])^2] \\ &= E[\mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{a}] \\ &= \mathbf{a}^\top E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] \mathbf{a} \\ &= \mathbf{a}^\top \text{Var}(\mathbf{X}) \mathbf{a}. \end{aligned}$$

By the same argument,

$$\text{Var}(Z) = \mathbf{c}^\top \text{Var}(\mathbf{X}) \mathbf{c}.$$

(b) Again using $\boldsymbol{\mu} = E[\mathbf{X}]$,

$$Y - E[Y] = \mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu}), \quad Z - E[Z] = \mathbf{c}^\top (\mathbf{X} - \boldsymbol{\mu}).$$

Hence

$$\begin{aligned} \text{Cov}(Y, Z) &= E[(Y - E[Y])(Z - E[Z])] \\ &= E[\mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{c}] \\ &= \mathbf{a}^\top E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] \mathbf{c} \\ &= \mathbf{a}^\top \text{Var}(\mathbf{X}) \mathbf{c}. \end{aligned}$$

(c) If $\mathbf{c} = \lambda \mathbf{a}$, then part (b) gives

$$\text{Cov}(Y, Z) = \mathbf{a}^\top \text{Var}(\mathbf{X}) (\lambda \mathbf{a}) = \lambda \mathbf{a}^\top \text{Var}(\mathbf{X}) \mathbf{a}.$$

From part (a),

$$\text{Var}(Y) = \mathbf{a}^\top \text{Var}(\mathbf{X}) \mathbf{a}, \quad \text{Var}(Z) = \lambda^2 \mathbf{a}^\top \text{Var}(\mathbf{X}) \mathbf{a}.$$

Therefore,

$$\begin{aligned} \text{Corr}(Y, Z) &= \frac{\text{Cov}(Y, Z)}{\sqrt{\text{Var}(Y)\text{Var}(Z)}} \\ &= \frac{\lambda \mathbf{a}^\top \text{Var}(\mathbf{X}) \mathbf{a}}{\sqrt{(\mathbf{a}^\top \text{Var}(\mathbf{X}) \mathbf{a})(\lambda^2 \mathbf{a}^\top \text{Var}(\mathbf{X}) \mathbf{a})}} \\ &= \frac{\lambda}{|\lambda|}. \end{aligned}$$

Hence

$$\text{Corr}(Y, Z) = \begin{cases} 1, & \lambda > 0, \\ -1, & \lambda < 0. \end{cases}$$

Bonus (10 points)

Let X and Y be random variables with finite variances, and assume that $\text{Var}(X) > 0$ and $\text{Var}(Y) > 0$.

(a) (4 points) Show that for any real number t ,

$$\text{Var}(X + tY) = \text{Var}(X) + 2t \text{Cov}(X, Y) + t^2 \text{Var}(Y).$$

(b) (6 points) Use part (a) to prove that

$$-1 \leq \text{Corr}(X, Y) \leq 1.$$

Solution.

(a) By definition,

$$\text{Var}(X + tY) = E\left[\left((X + tY) - E[X + tY]\right)^2\right].$$

Since expectation is linear,

$$E[X + tY] = E[X] + tE[Y].$$

Therefore,

$$(X + tY) - E[X + tY] = (X - E[X]) + t(Y - E[Y]).$$

Expanding the square,

$$\begin{aligned}\text{Var}(X + tY) &= E\left[\left((X - E[X]) + t(Y - E[Y])\right)^2\right] \\ &= E[(X - E[X])^2] + 2tE[(X - E[X])(Y - E[Y])] + t^2E[(Y - E[Y])^2] \\ &= \text{Var}(X) + 2t \text{Cov}(X, Y) + t^2 \text{Var}(Y).\end{aligned}$$

Hence

$$\boxed{\text{Var}(X + tY) = \text{Var}(X) + 2t \text{Cov}(X, Y) + t^2 \text{Var}(Y).}$$

(b) For any real number t ,

$$\text{Var}(X + tY) \geq 0.$$

By part (a),

$$\text{Var}(X + tY) = \text{Var}(Y)t^2 + 2 \text{Cov}(X, Y)t + \text{Var}(X).$$

This quadratic in t is nonnegative for all real t , so its discriminant must be non-positive. Therefore,

$$(2 \text{Cov}(X, Y))^2 - 4 \text{Var}(X) \text{Var}(Y) \leq 0.$$

Hence

$$\text{Cov}(X, Y)^2 \leq \text{Var}(X) \text{Var}(Y).$$

Dividing by the positive quantity $\text{Var}(X) \text{Var}(Y)$ gives

$$\frac{\text{Cov}(X, Y)^2}{\text{Var}(X) \text{Var}(Y)} \leq 1.$$

By definition of correlation,

$$\text{Corr}(X, Y)^2 \leq 1.$$

Therefore,

$$\boxed{-1 \leq \text{Corr}(X, Y) \leq 1.}$$